

# Working with Virtual Machines

Virtual Machines (VMs) are **complete computers in software** – with virtual CPU, memory, hard drives, and network cards. A **hypervisor** like VirtualBox, VMware, or Hyper-V provides this hardware.

## The Host

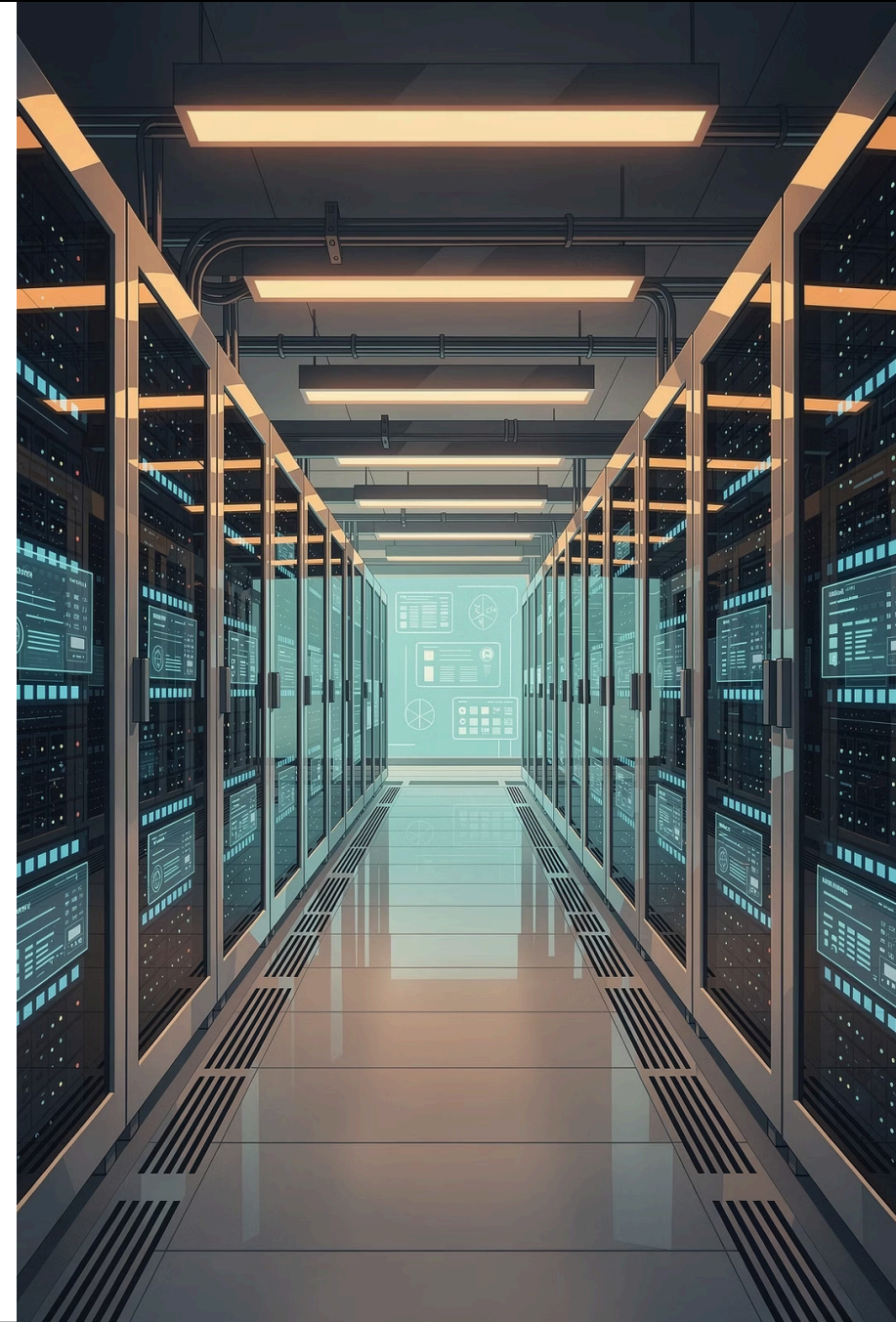
Your actual, physical computer

## The Guest

The virtual machine

## The VM

State, configuration, and virtual disks



## CHAPTER 1

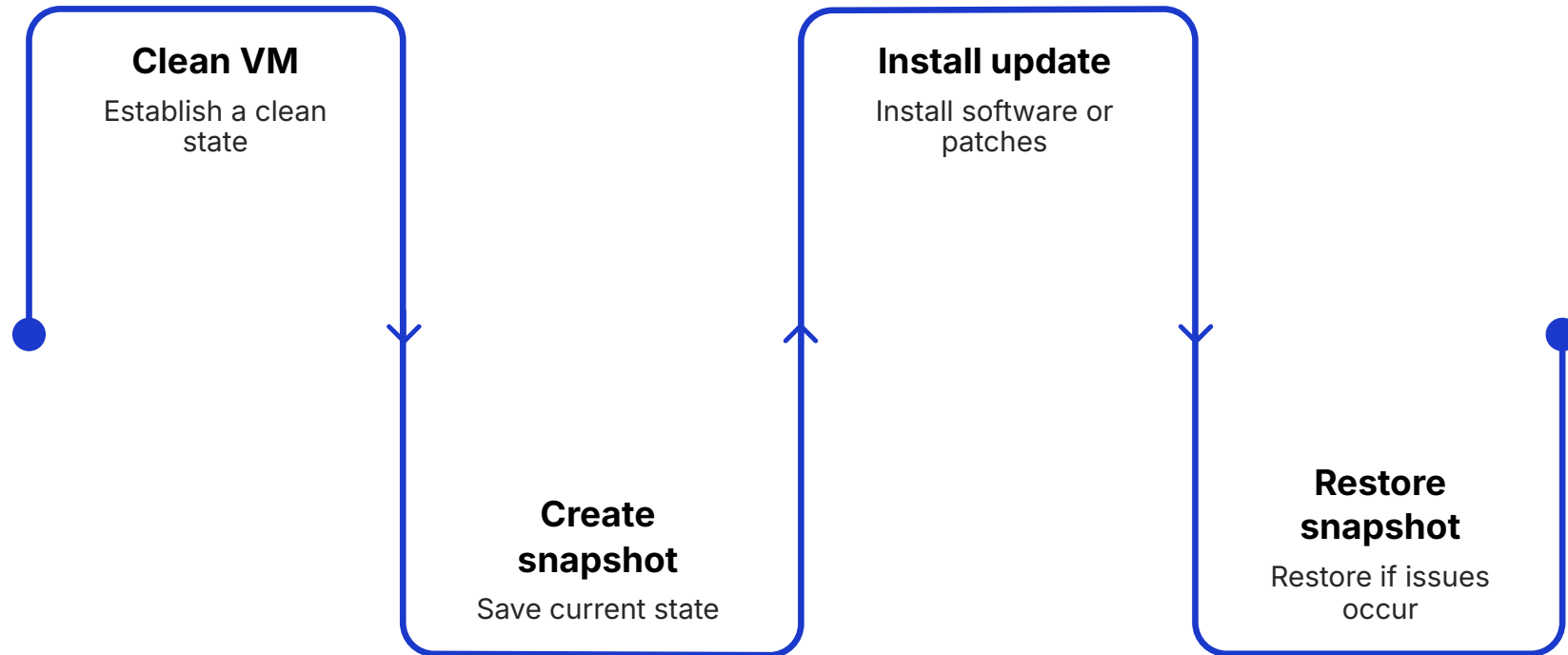
# Snapshots: Fundamentals

A **snapshot** is a **restore point** for a VM. It captures the state of the virtual hard disks, the configuration, and optionally the system memory. In Hyper-V, this concept is called a **checkpoint**.

Technically, after a snapshot is taken, **changes are written to differencing files** – the original state remains intact. This allows you to revert to a previous point in time at any moment.

## CHAPTER 1

# Snapshots in Practice



The big advantage: You can **experiment quickly** without having to completely reset the VM. Snapshots enable stress-free learning and testing.

## CHAPTER 1

# When are snapshots useful?



## Before Updates

Before major Windows or kernel updates as a safety net



## Before Configuration Changes

Network, firewall, user permissions, services



## Before Installations

Databases, web servers, drivers, or development environments



## Before Experiments

Security tests, automation, malware analysis

## CHAPTER 1

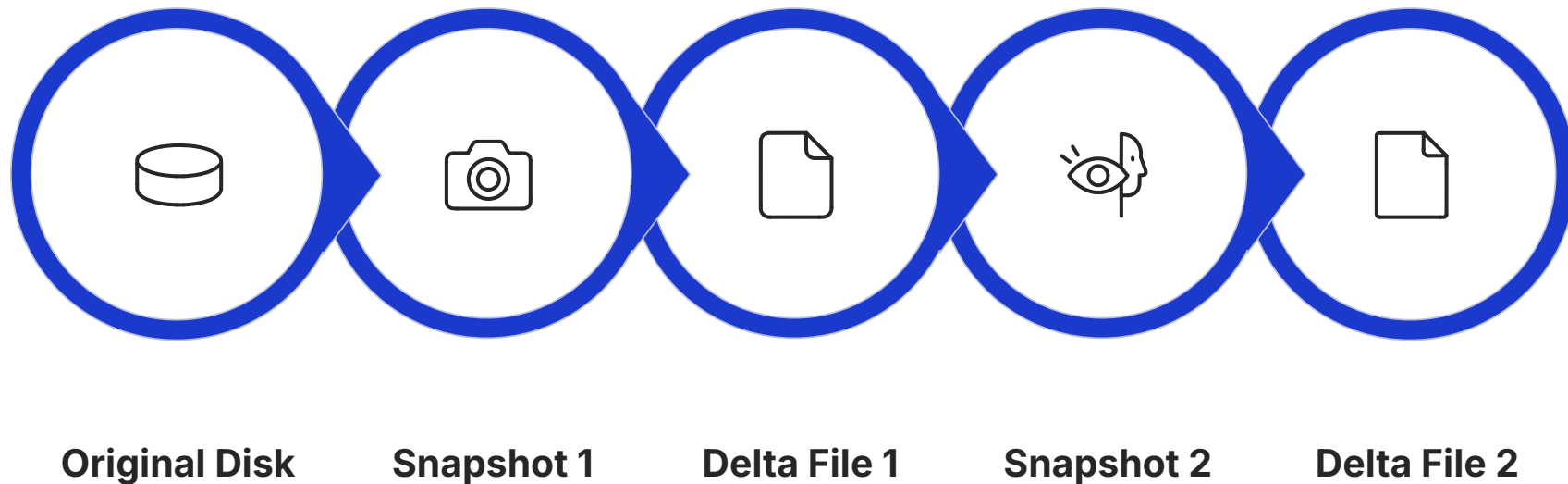
# What is saved in a snapshot?

| Component        | Saved?   | Significance                           |
|------------------|----------|----------------------------------------|
| Virtual Disk     | Yes      | Files and data on the VM               |
| VM Configuration | Yes      | RAM allocation, devices, network cards |
| Memory Contents  | Optional | VM continues exactly where it left off |
| Running State    | Often    | Whether the VM was on, off, or paused  |

If the memory is also saved, the recovery acts like a **frozen and thawed computer**.

## CHAPTER 1

# How Snapshots Work Internally



The original state is preserved, and new changes are stored in additional files. This has consequences: **increased storage requirements**, more complexity, and potentially **worse performance** with long snapshot chains.

## CHAPTER 1

# Snapshots ≠ Backups

**Key Takeaway: Snapshot = point-in-time recovery. Backup = true data protection.**

| Snapshot                           | Backup                         |
|------------------------------------|--------------------------------|
| Fast recovery point                | Full data protection           |
| For tests and short-term rollbacks | For failure, loss, restoration |
| Within the same VM structure       | Ideally stored separately      |
| Not for long-term retention        | Specifically designed for that |

Microsoft explicitly points out for Hyper-V: **Checkpoints are not a substitute for a backup.**

## CHAPTER 1

# Common Snapshot Mistakes

## Keeping too many snapshots

Long chains make the VM messy and degrade performance and storage consumption.

## Treating a snapshot as a backup

One of the most dangerous misconceptions of all.

## Using blindly on databases

For services with heavy write activity, consistency must be ensured.

## Working without a naming convention

Better: *Before Apache installation 2026-03-24* instead of "Snapshot 1" or "Test".

## CHAPTER 1

# Best Practices for Snapshots

01

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## Create snapshots before risky changes

Never retrospectively—the timing is crucial.

03

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## Remove unnecessary snapshots

Clean up after a successful test; avoid endless chains.

02

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## Use short, clear descriptions

Record the date and reason in the name.

04

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## Export for actual backup

Keep them separate: test point, copy, archive, and backup.

## CHAPTER 2

# Cloning Virtual Machines

When **cloning**, you create another VM from an existing one. This is based on the original but runs as its own system afterwards. Typical use cases:

- Creating multiple test systems from a base VM
- Distributing a prepared training environment
- Golden base system for various roles
- Building a lab network with multiple similar machines



## CHAPTER 2

# Full vs. Linked Clone

## Full Clone

A complete, independent copy including virtual disks. No longer depends on the original.

- Independent and stable
- Good for long-term use
- Good for distribution and archiving
- Requires more storage space

## Linked Clone

Uses the original or snapshot as a common base. Only stores changes separately.

- Quick to create, saves storage
- Good for short-term testing
- Depends on the original
- Less robust for long-term use

 **Note:** Full clone = own complete machine. Linked clone = derived machine with dependency.

## CHAPTER 2

# Pitfalls when Cloning

A cloned VM is **technically copied**, but often not yet properly individualized from an administrative standpoint. These identities must be checked or re-generated:

Hostname

IP Address

MAC Address

Machine IDs

SSH Host Keys

Certificates &amp; Tokens

Two cloned systems with identical configuration in the same network can lead to **conflicts** – especially with duplicate IP addresses or identical system identifiers.

## CHAPTER 2

# Importing and Exporting

An **export** packages a VM into a transportable file. An **import** reads this package into a hypervisor. The **OVF/OVA** format is very commonly used – a standardized format for packaging and distributing virtual appliances (DMTF).

## OVF

Description plus separate accompanying files

## OVA

Single archive that bundles everything together



CHAPTER 2

# Snapshot, Clone, Export – The Differences

| Function | Main Purpose                         | Typical Use Case                |
|----------|--------------------------------------|---------------------------------|
| Snapshot | Save state and revert to it          | Before changes and tests        |
| Clone    | Create a new VM from an existing one | Labs, multi-system environments |
| Export   | Transport or archive a VM            | Distribution, migration         |
| Import   | Make an exported VM usable again     | Transfer to another host        |

**Snapshot** = Revert point · **Clone** = Additional VM · **Export** = Transport · **Import** = Commissioning

## CHAPTER 2

# Security during export: What travels with it?

Before exporting, you should check whether the VM contains **sensitive data**. Often, the following are exported unintentionally:

**Passwords &  
API keys**

**SSH keys &  
certificates**

**Browser data &  
tokens**

**Personal data**

An export is therefore not just a technical matter, but also a **security and data protection issue**. Clean up the VM before sharing: check for temporary files, keys, logs, hostnames, and network configuration.



## CHAPTER 3

# Virtual Networks

A VM is only truly usable when it can communicate with other systems. A VM has one or more **virtual network cards** that the hypervisor connects to a virtual network.

The three most important network modes are:

**NAT**

Network Address Translation

**Bridged**

Network Bridge

**Host-only**

Host-only network



## CHAPTER 3

# NAT – Network Address Translation

## How it works

The VM resides in a private virtual network. The host translates connections to the outside – similar to a home router. Private address ranges according to RFC 1918 (e.g., 192.168.0.0/16).

## Typical use case

Installing software, downloading updates, browsing the web – without the VM being directly visible in the LAN.

## Characteristics

- Outgoing internet access: ✓
- Directly visible in LAN: ✗
- Incoming connections: only via port forwarding

## CHAPTER 3

# Bridged and Host-only



## Bridged

The VM appears on the LAN like its **own physical computer** with its own MAC and IP address. Ideal for server services, SSH, web servers, or client/server scenarios. Higher visibility, however, also means a larger attack surface.



## Host-only

Host and VM communicate with each other without the VM having direct access to the external network. Ideal for **isolated learning environments**, malware labs, and secure experiments without LAN exposure.

## CHAPTER 3

# Direct Comparison of the Three Modes

| Mode      | Internet Access | Visible in LAN | Host → VM | Good for                  |
|-----------|-----------------|----------------|-----------|---------------------------|
| NAT       | Yes (outbound)  | Mostly no      | Limited   | Standard lab, updates     |
| Bridged   | Yes             | Yes            | Yes       | Realistic network testing |
| Host-only | No by default   | No             | Yes       | Isolated labs             |

❏ **Basic principle:** Always choose the least open network mode that still meets your goal – **Least Exposure**.

## CHAPTER 3

# Understanding Communication Between VMs

Network communication rarely fails due to a single point, but often because of a **chain of conditions**. These questions must all be answered:

01

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**Correct network mode?**

Are both systems connected in the appropriate mode?

02

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**Valid IP addresses?**

Do both systems have correct addresses?

03

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**Same subnet or routing?**

Can the packets find their way?

04

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**Firewall allows communication?**

Are the required ports enabled?

05

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**Service actually running?**

Is the desired service active on the target system?

## CHAPTER 3

# Communication across multiple layers

| Layer            | Question                                            |
|------------------|-----------------------------------------------------|
| Physical/Virtual | Is the network adapter connected?                   |
| Layer 2          | Are the systems on the appropriate virtual network? |
| Layer 3          | Do they have IP addresses and routing?              |
| Layer 4          | Is the required port open?                          |
| Application      | Is the service actually running?                    |

## Example: VM A (Web Server) → VM B (Client)

1. Both on the same network
2. Both have valid IP addresses
3. VM B knows the destination address of VM A
4. Firewall on VM A allows the web port
5. Web server is actually running on VM A

Only then does `http://IP-of-VM-A` work.

## CHAPTER 3

# Why ping sometimes doesn't work

Ping uses **ICMP**, which can be blocked by firewalls. The destination might be reachable, but it does not respond to ping.

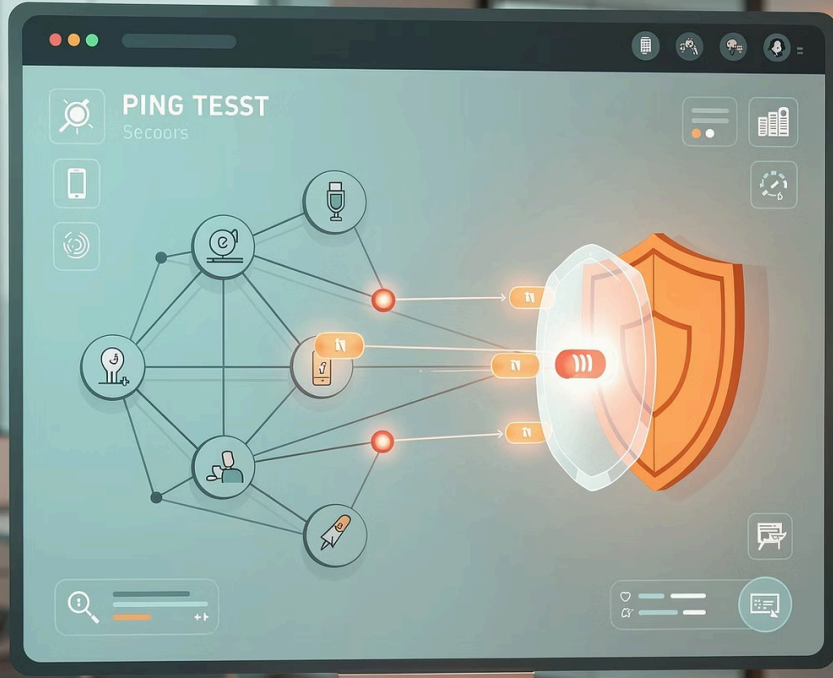
## Ping successful

Connection is very likely present

## Ping fails

Not automatically dead – maybe just ICMP blocked

Additionally, check: IP configuration, routing, DNS, open ports, firewall rules, and actually running services.



## CHAPTER 4

# Security aspects of virtual networks

Virtual networks are not automatically secure just because they are virtual. **Isolation is possible, but must be consciously configured.** You decide how visible and how accessible a VM is.

## NAT

Good default choice: VM has internet access, but is not openly visible on the LAN

## Bridged

Higher visibility and thus a larger attack surface on the LAN

## Host-only

Good isolation for testing between host and VM



## CHAPTER 4

# Port Forwarding and Clones: Security Risks

## Use Port Forwarding Sparingly

With NAT, individual services can be made accessible from the outside. Best practice:

- Only open truly necessary ports
- Only for as long as needed
- Document and remove after testing

## Cloned VMs: What secrets travel with them?

- Identical SSH host keys
- Stored admin logins
- API tokens and certificates
- Old logs containing sensitive data
- Static IP addresses and hostnames

## CHAPTER 4

# Principle of Least Privilege

Always choose the least open network mode that still accomplishes your goal.

1

**Download updates only**

→ NAT

2

**Local testing with host**

→ Host-only

3

**LAN access to VM**

→ Bridged

This principle is known as **Least Exposure** – minimizing the attack surface. Additionally: Keep the guest operating system up to date, disable unnecessary services, enable the firewall, avoid default passwords, and only activate shared folders when needed.

# The Right Learning Model

## Snapshot

I want to be able to roll back.

## Clone

I want another VM based on this one.

## Export

I want to transport or share the VM.

## Import

I want to put such a package back into operation.

Security in virtual networks is primarily created through conscious visibility, clear boundaries, and minimal permissions.

